# Ali Al-haj

**Date of birth:** 20/05/1994 | **Nationality:** Syrian | **Gender:** Male | **Phone number:** (+91) 9063998307 (Mobile) | **Email address:** ali.n.alhag@gmail.com | **LinkedIn:** https://www.linkedin.com/in/ali-alhaj-a62617232/ |

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**Address:** 500019, HYDERABAD, India (Hyderabad, India.) | **Address:** Tartous, Syria (Home)

## ABOUT ME

Highly motivated researcher with a Ph.D. in Computer Science, specializing in SDN, programmable networks (P4), network security, and cloud/virtualized environments. Experienced in building advanced network testbeds, deploying virtualized and cloud-based systems, and publishing in top venues (ICDCN, AEU, Computer Communications, etc.). Skilled in VMware, Linux administration, containerization, and cloud platforms, with a strong background in student mentoring and collaboration on large-scale networking projects.

## EDUCATION AND TRAINING

30/08/2021 – CURRENT Hyderabad, India

### PHD IN COMPUTER SCIENCE University Of Hyderabad

Develop P4-Enable switch solutions to improve SDN network security.

**Website** https://uohyd.ac.in/ | **Field of study** Network security

06/09/2019 – 30/05/2021 RAJKOT, Gujarat,, India

### MASTER DEGREE Marwadi university

Completed Master degree in Cyber Security under the supervision of Dr.Nitul Dutta

**Address** 360006, RAJKOT, Gujarat,, India | **Website** https://www.marwadiuniversity.ac.in/ |

**Field of study** Computer Engineering- Cyber security | **Final grade** 8.66 |

**Thesis** Design and Analyze a Secure Architecture for SDN Network

01/09/2012 – 18/03/2018 Latakia, Syria

### BACHELOR DEGREE Tishreen University

graduated from the Faculty of Informatics Engineering at Tishreen University, Department of Computer Systems and Networks, with a GPA of 87.66%, and with the rank of first graduate for the year 2017.

**Website** https://tishreen.edu.sy/en | **Field of study** informatics Engineering- Department of Computer Systems and Networks |

**Final grade** 87.66%

## PUBLICATIONS

2026

### Efficient deployment of programmable switches for scalable network monitoring

(SCIE, Scopus) (Q1)

Ali Nadim Alhaj, Wilson Naik Bhukya, Rajendra Prasad Lal

**Journal Name**: Computer Communications

2025

### Adaptive network-wide superspreader detection using programmable switches

Alhaj, A. N., Bhukya, W. N., & Lal, R. P. (2025). Adaptive network-wide superspreader detection using programmable switches. AEU-International Journal of Electronics and Communications, 156041.

2025

### Advanced Method for Flow Frequency Estimation and Top-k Heavy Hitters Detection in SDN Leveraging Programmable

In 26th International Conference on Distributed Computing and Networking (ICDCN 2025),
https://doi.org/10.1145/3700838.3700853
(Ranking B conferance)

Ali Alhaj, Dr.Wilson Naik, Dr. RP lal.

2024

### A Novel Space-Efficient Method for Detecting Network-Wide Heavy Hitters in Software-Defined Networking Using P4-Switch

Research paper in title - A Novel Space-Efficient Method for Detecting Network-Wide Heavy Hitters in Software-Defined Networking Using P4-Switch**.**
in International Arab Journal of Information Technology (SCIE, Scopus (Q2)). sep-2024.

Ali Alhaj, Dr.Wilson Naik, Dr. RP lal.

2024

### A Space-Saving Technique in SDN for Identifying Top-k Super-Spreaders Using P4-Enabled Switches.

Alhaj, A. N., Bhukya, W. N., & Lal, R. P. (2024, October). A Space-Saving Technique in SDN for Identifying Top-k Super-Spreaders Using P4-Enabled Switches. In International Conference on Computing and Communication Networks (pp. 659-673).

2023

### Design and Analysis of a Robust Security Layer for Software Defined Network Framework

Alhaj, Ali Nadim, et al. "Design and analysis of a robust security layer for software defined network framework." *International Journal of Sensor Networks* 46.1 (2024): 1-14.

Ali Nadim Alhaj; Narottam Das Patel; Ajeet Singh; Rohit Kumar Bondugula; Mohsin Furkh Dar; Jameel Ahamed

2022

### Analysis of Security Attacks in SDN Network: A Comprehensive Survey

published in International Conference on Communication, Cloud, and Big Data (CCB) 2020
18-19 December 2020.

Ali Alhaj, Netul Dutta

## PATENTS

### An Artificial Intelligence (AI)-Based Legal Decision Support System for Refugee Status Determination and Method Thereof

An Artificial Intelligence (AI)-Based Legal Decision Support System for Refugee Status Determination and Method Thereof, Indian Patent Application No. 202541079153 A, published 29 August 2025.
Inventors: Ali Nadim Alhaj, Basel Abou Rokba, Mohammad Ramin Hakimy, Prof. Dr. S. Sumitra.

## WORK EXPERIENCE

**RESEARCH SCHOLAR/ TEACHER ASSISTANT** – 18/10/2021 – Current – HYDERABAD, INDIA

- Conduct advanced research in **SDN, P4-based programmable networks, traffic measurement, and network security**
- Build and manage **virtualized/cloud-based SDN testbeds** used for experiments
- Configure and maintain **Linux servers, containers, and virtual machines** for research projects
- Publish research results in **top venues** (Computer Communications, ICDCN, IAJIT, AEU, etc.)

- Mentor undergraduate and postgraduate students in networking, systems, and security
- Working with SmartNic/ FPGA

**FRONT-END DEVELOPER – OMEGA CROP, ALI AL BAKER TEAM** – 01/11/2020 – 01/08/2023 – ONLINE, UNITED KINGDOM

Working online with a team of programmers in the UK - To develop a huge site to help supervise the agricultural seasons for farmers and monitor productivity and weather factors based on Deck.gl maps.
Project URL: https://www.omegacrop.com/

**TEACHER ASSISTANT – TISHREEN UNIVERSITY** – 01/09/2018 – 04/09/2019 – LATAKIA, SYRIA

Assistant Lecturer in the College of Informatics Engineering - Department of Computer Systems and Networks - Teacher for Network, Systems and Computer Modeling Courses

## LANGUAGE SKILLS

Mother tongue(s): **ARABIC**

Other language(s):

| | UNDERSTANDING | | SPEAKING | | WRITING |
|---|---|---|---|---|---|
| | Listening | Reading | Spoken production | Spoken interaction | |
| **ENGLISH** | C1 | C1 | C1 | C1 | C1 |

*Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user*

## SKILLS

python | Front-end Developer | Networking | C, C++, C# | Java Programming language | Web Development: HTML 5, CSS, JavaScript | GIT (GitHub) | Experienced in React Js frameworks | SmartNIC | FPGA

**Core Skilled**

Virtualization (VMware vSphere VMWare vSAN) | Operative systems LINUX | research and analytics skills | Docker | Strong communication and problem-solving skills | SDN

## RECOMMENDATIONS

**Prof. Siba K.Udgata** Professor

Professor and Dean, School of Computer and Information Sciences, University of Hyderabad

**Email** udgata@uohyd.ac.in | **Phone** (+91) 9705395136

**Dr. Wilson Naik** Associate Professor

Associate Professor, School of Computer & Information Sciences, University of Hyderabad, India.

**Email** rathore@uohyd.ac.in | **Phone** (+91) 9493144443

**Dr. Rajendra Prasad Lal** Assistant Professor

School of Computer & Information Sciences, University of Hyderabad, India.

**Email** rajendraprasd@uohyd.ac.in | **Phone** (+91) 7893937026